

Lecture 6

Measures of Spread

Review

1 One value is far larger than all the rest. Which moves more, the mean or the median?

The mean, which gets pulled toward it. The median barely moves – it only cares about position in the ordered list.

2 We wrote the mean three different ways last time. What were they?

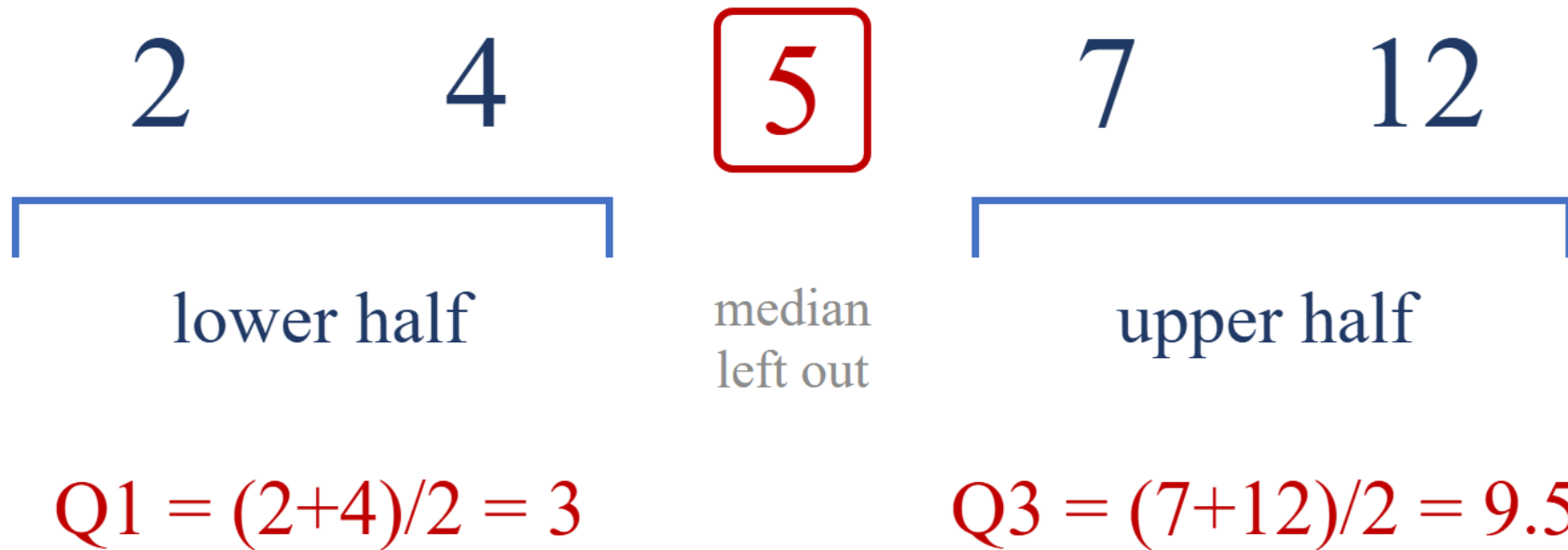
Add the values and divide by n ; or work from a frequency table; or from a relative frequency table. All three give the same number.

3 A student's exam score is at the 75th percentile. What does that tell you?

That 75 percent of the scores are at or below theirs.

A note on quartiles: which definition?

- There are several accepted ways to calculate a quartile. In this class we use the **median of halves** method.
- Find the median first. When n is odd, leave that median out when forming the lower and upper halves.



Why R may give you different quartiles

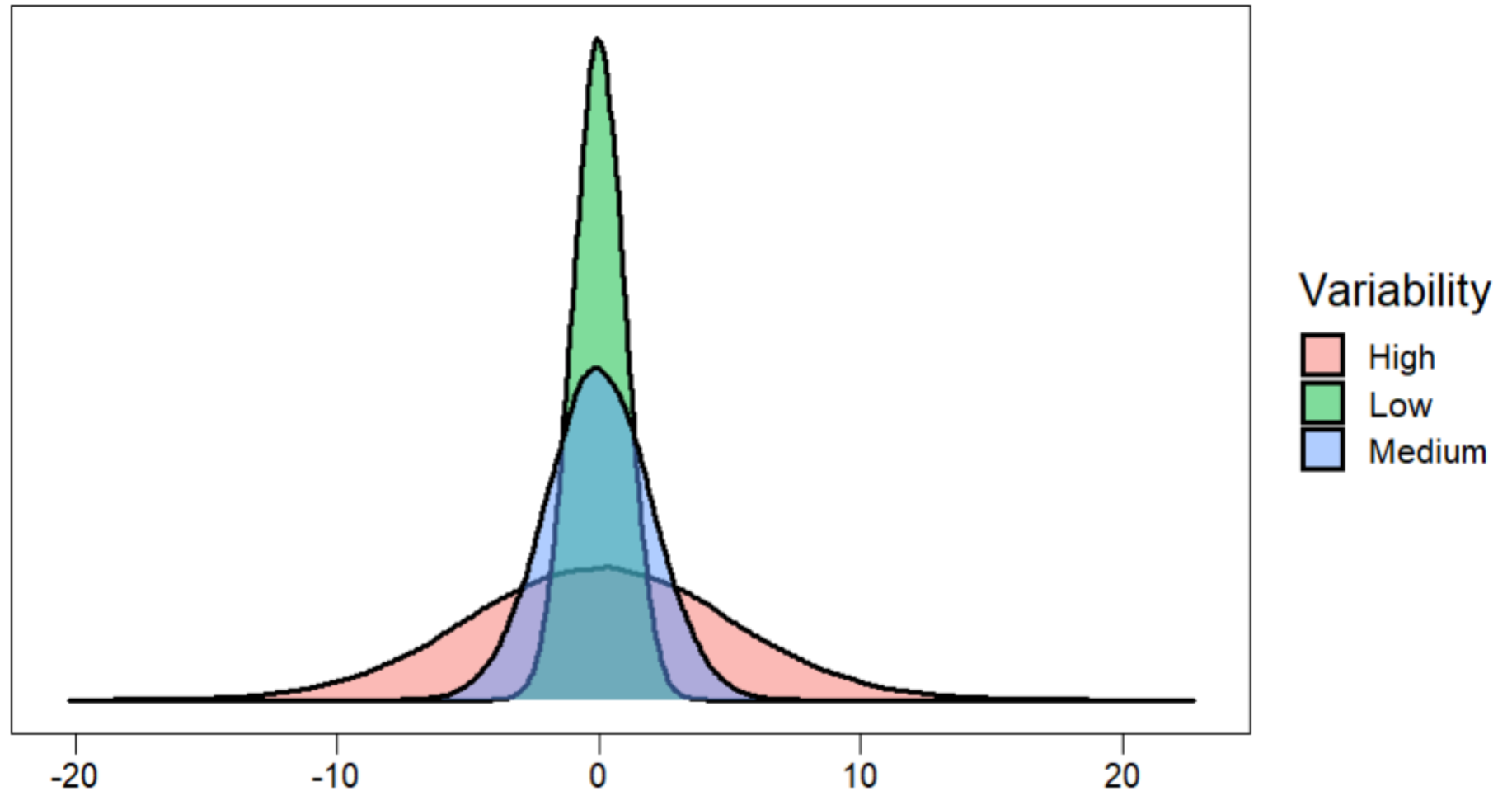
Method	Q1	Median	Q3
Median of halves – this class	3	5	9.5
R quantile() default	4	5	7
R boxplot() hinges	4	5	7

- R's quantile() interpolates between the ordered values; R's boxplot() is different again, drawing the box at Tukey's hinges, which keep the median inside both halves
- A disagreement does not mean either of you made a mistake - it means a different definition was used
- In this class, use the median of halves, and apply the same method **consistently**

Our running example for the rest of today

- We will build every measure of spread from the same five values, on the board
 - $x = 2, 4, 5, 7, 12$ $n = 5$
 - $\text{mean} = (2 + 4 + 5 + 7 + 12) / 5 = 30 / 5 = 6$
- and we already have its quartiles, from the slide before:
 - $Q1 = 3$ $Q2 = 5$ $Q3 = 9.5$
- Still to come: the range, the IQR, the variance and standard deviation, and the boxplot
- Keep these numbers - everything from here on is worked from them

Variability of A Distribution: Measures of Spread

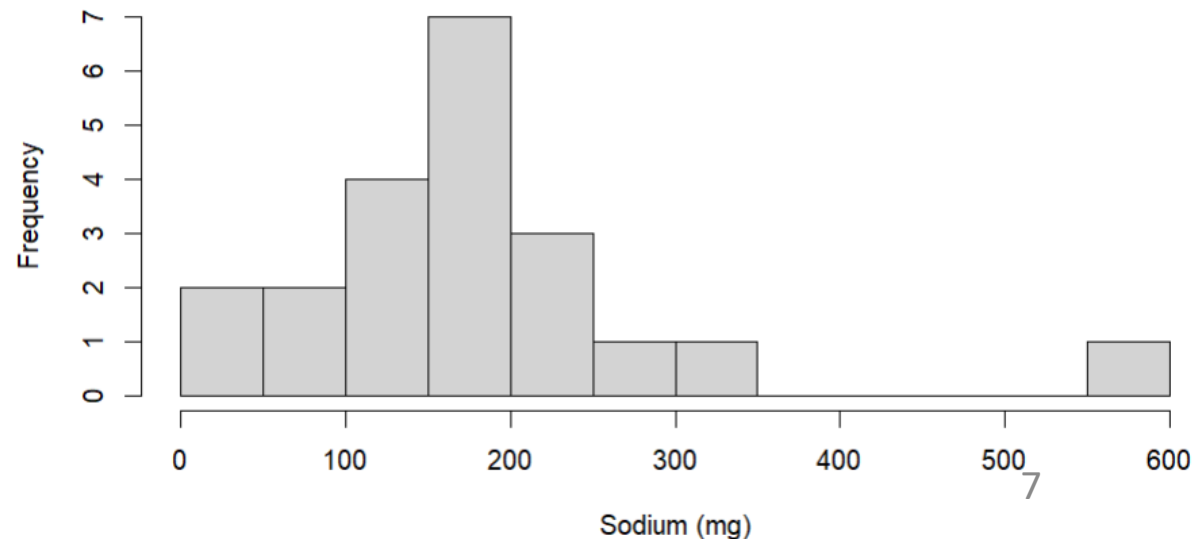
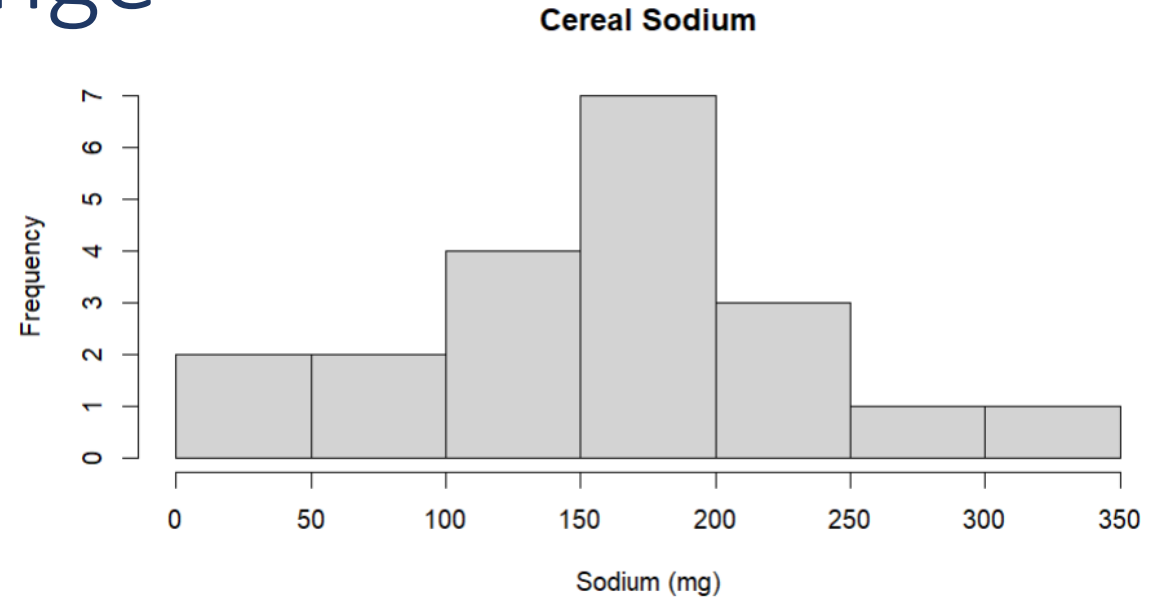


Measures of Spread: Range

- The **range** is a measure of the distance between the smallest and largest values in the data

The range can be computed with only two data points the minimum value and maximum value

- If the range of a set of data is large, then usually this indicates greater dispersion of values
- The range is severely affected by the presence of outliers
- We typically do not use the range to measure variability



Measures of Spread: Interquartile Range

- The **interquartile range (IQR)** measures the spread of the middle 50% of the observations
- It is resistant to outliers

$$IQR = Q3 - Q1$$

- The more variability the larger the value of the **IQR**
- **IQR** is a good choice for distributions that are highly skewed!

Practice: Finding quartiles and IQR

$x = 2, 4, 5, 7, 12$

Compute the range, and then the IQR

Working:

$$\text{range} = \text{max} - \text{min} = 12 - 2 = 10$$

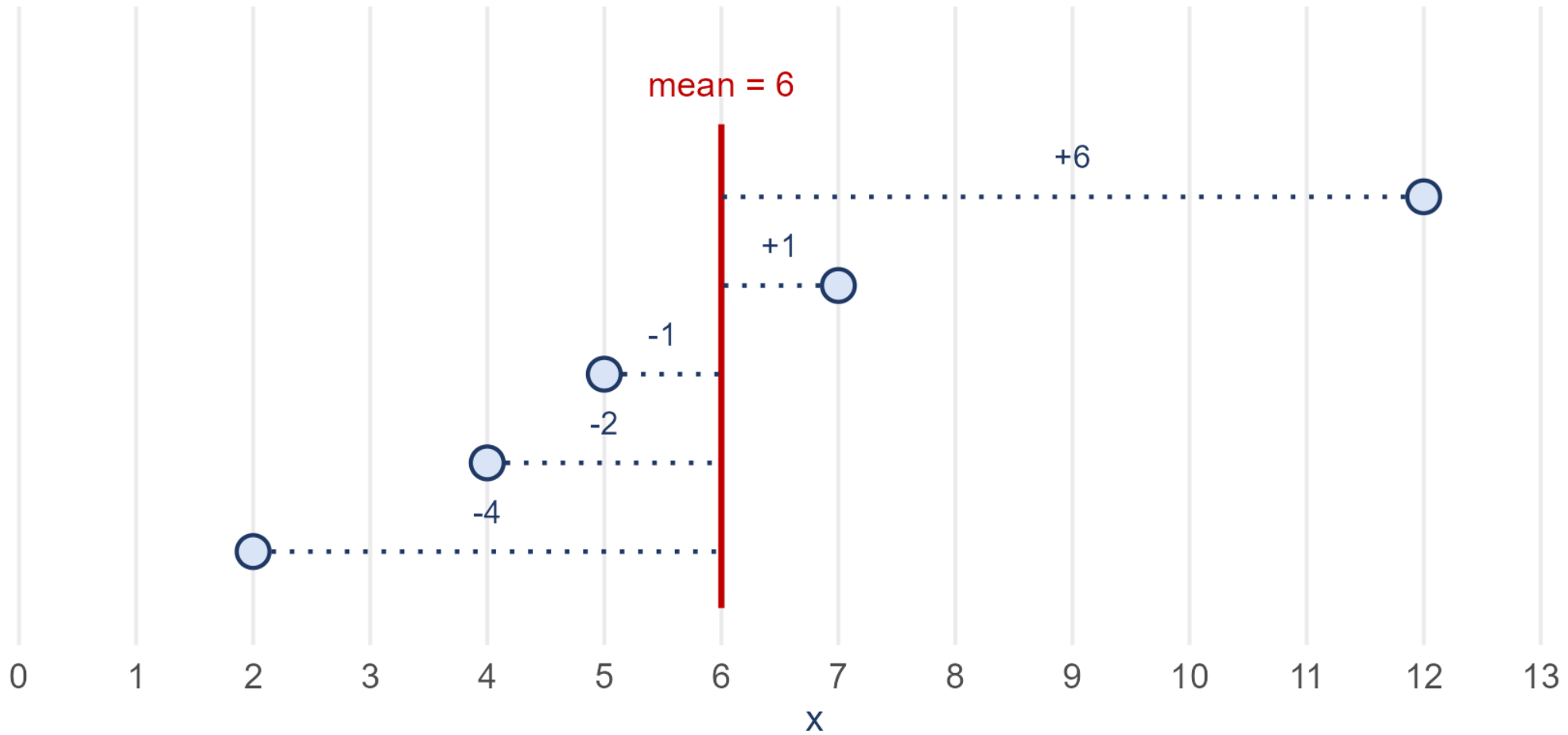
$$Q1 = 3 \quad Q3 = 9.5 \quad (\text{from the running example})$$

$$\text{IQR} = Q3 - Q1 = 9.5 - 3 = 6.5$$

The middle half spans 6.5 units; the full range spans 10.

Measures of Spread: Deviation

- A better measure of variability that uses *all* the data is based on **deviations**
- **deviations** are the distances of each value from the mean of the data:
Deviation of an observation $x_i = (x_i - \bar{x})$
- Every observation will have a deviation from the mean



Each deviation is the gap from a value to the mean. Three are negative, two positive - and they cancel exactly.

The deviations always sum to zero

- Our five cancelled exactly. That is not luck - it happens for every dataset:
 - $\Sigma(x - \text{mean}) = \Sigma x - n \cdot \text{mean}$
 - but $\text{mean} = \Sigma x / n$, so $n \cdot \text{mean} = \Sigma x$
 - **hence** $\Sigma(x - \text{mean}) = \Sigma x - \Sigma x = 0$
- Check it on our example: $-4 - 2 - 1 + 1 + 6 = 0$
- So averaging the deviations is useless - it reports 0 for tightly packed data and for wildly scattered data alike
- We have to remove the signs before averaging. That is why the next step squares them.

Measures of Spread: Variance

- The sum of all deviations is zero. $\sum_{i=1}^n (x_i - \bar{x}) = 0$
- We typically use either the **squared deviations** or their **absolute value**
Squared deviation of an observation $x_i = (x_i - \bar{x})^2$
- The **Variance** of a distribution is the average squared deviation from the mean

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sum $\sum_{i=1}^n (x_i - \bar{x})^2$ is called the sum of squares

Measures of Spread: Standard Deviation

- Since the variance uses the squared deviation, we usually take its square root called the **standard deviation**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- The standard deviation represents (roughly) the average distance of an observation from the mean
- The greater s is the greater the variability in the data is
- We denote the population parameter for the variance and standard deviation using σ for s and σ^2 for s^2

Why divide by $n - 1$?

- We divide by $n - 1$ because we have only $n - 1$ pieces of independent information for s^2
- Since the sum of the deviations must add to zero, then if we know the first $n - 1$ deviations we can always figure out the last one
- Ex.) suppose we have two data points and deviation of the first data point is $x - \bar{x} = -5$
 - Then the deviation of the second data point has to be 5 for the sum of deviations to be zero.

Try it out: computing the variance and standard deviation

- $x = 2, 4, 5, 7, 12$
- $\text{mean} = 6$
- Find the deviations, then the sum of squares, then the variance and the standard deviation

Working:

deviations -4 -2 -1 1 6

(sum 0)

squared 16 4 1 1 36

sum of squares = 58

$$s^2 = 58 / (5 - 1) = 14.5$$

$$s = \text{sqrt}(14.5) = 3.81$$

s is in the units of x ; the variance is in those units squared.

Your s is itself an estimate

- Every number today describes the one sample you happened to collect. Roll ten more times and all of them move.
- That movement from sample to sample is called **sampling variation**

	ten rolls	mean	s
our sample	1 2 3 3 4 4 4 5 6 6	3.80	1.62
sample 2	2 2 3 3 3 3 4 4 6 6	3.60	1.43
sample 3	1 2 2 3 3 4 4 4 5 6	3.40	1.51
sample 4	1 1 3 4 4 4 5 6 6 6	4.00	1.89
sample 5	1 1 2 3 4 4 6 6 6 6	3.90	2.08

How much does the mean move?

Its standard deviation across repeated samples is called the **standard error**:

$$\begin{aligned} \text{SE} &= s / \sqrt{n} \\ &= 1.619 / \sqrt{10} \\ &= 0.512 \end{aligned}$$

Bigger n, smaller SE.

Not on Exam 1 - this is the idea behind margin of error, which we pick up in a few weeks.

Which measures do you report?

- A center on its own does not describe a distribution - it needs a spread beside it, and the two come in matched pairs
 - the mean and the standard deviation – both use every observation, so both are pulled by an outlier
 - the median and the IQR – both are positions in the ordered data, so both are resistant
- A measure is **resistant** when a single extreme observation barely moves it
- Roughly symmetric, no outliers: report the mean and the standard deviation
- Skewed, or outliers present: report the median and the IQR
- Do not mix the pair. A mean next to an IQR describes two different things and answers neither question well.

The five number summary

- **Five numbers** are enough to sketch a whole distribution

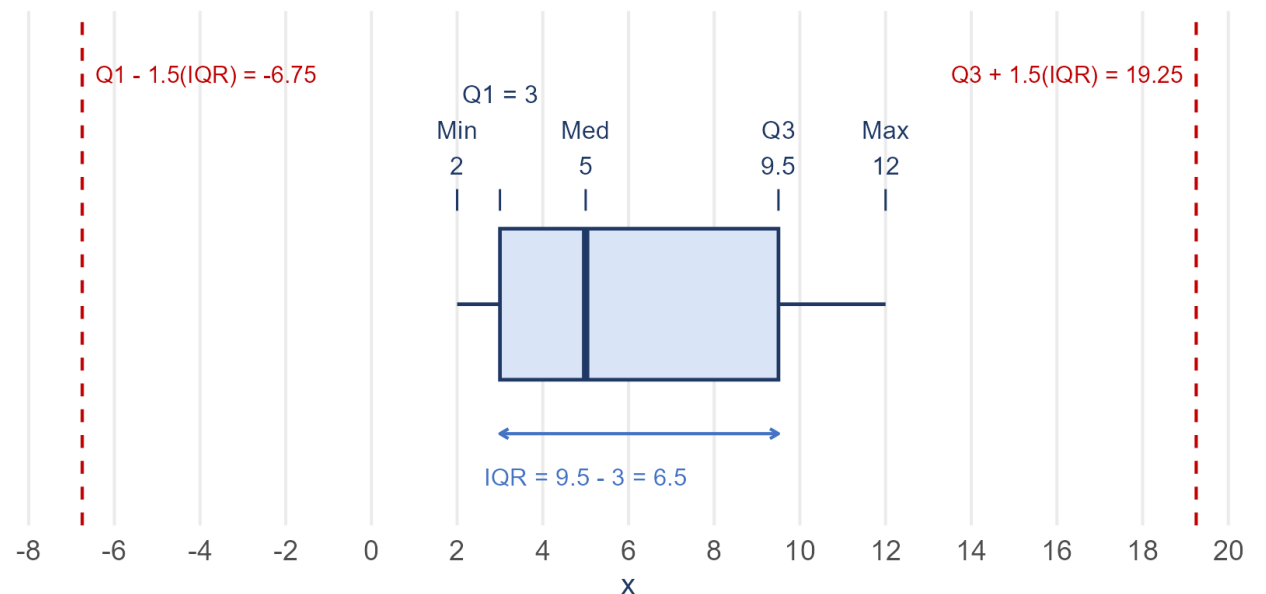
- **Minimum Q1 Median Q3 Maximum**

- **For $x = 2, 4, 5, 7, 12$: 2 3 5 9.5 12**

- the median gives the center
- Q1 and Q3 bracket the middle half, and their difference is the IQR
- the minimum and maximum give the full extent
- These five numbers are exactly what a boxplot draws

The Boxplot (Box and Whisker Plot): A five number summary

- Pros:
 - good for describing shape and location
 - Can be used to identify outliers
 - Length of whiskers indicates skew
 - Good for comparing two distributions or across categories
- Cons:
 - does not show certain features like mounds, or gaps as well as a histogram



Identifying outliers: the 1.5 x IQR rule

- We need a rule for "extreme" that does not depend on eyeballing the plot
- The **fences** sit an extra one and a half IQRs beyond the box:
 - $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$
- Any observation outside a fence is an **outlier**, and gets plotted as its own point
- The whiskers then stop at the most extreme observation INSIDE the fences - not at the minimum and maximum
- The fences themselves are never drawn on the plot
- Our example: $3 - 1.5(6.5) = -6.75$ and $9.5 + 1.5(6.5) = 19.25$, so nothing is flagged and the whiskers reach 2 and 12
- Being flagged does not make an observation wrong. It makes it worth a second look.
- Careful: R's `boxplot()` would draw this box at 4 and 7, not 3 and 9.5 - it uses Tukey's hinges, the third row of the table back on slide 4

Ex. Construct a Boxplot

- Consider the following data which come from 20 rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2 , 2, 2, 2, 2, 3		3, 3, 3, 3, 4, 5 , 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

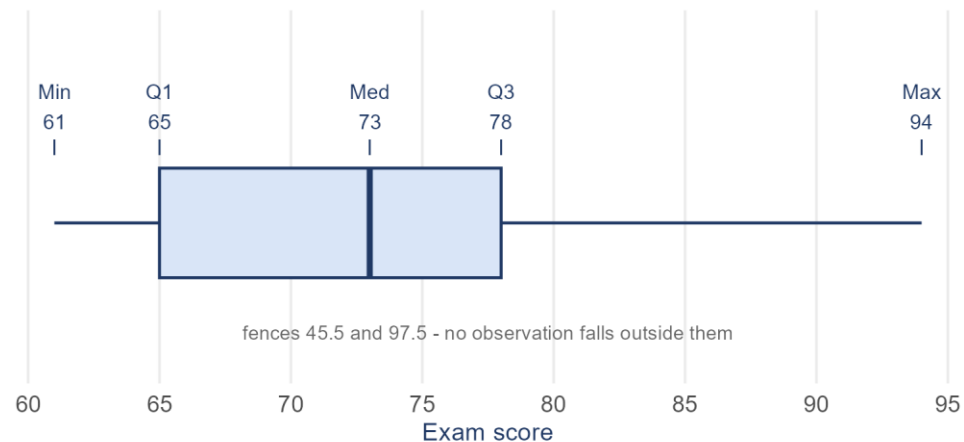
$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Practice: Construct a Boxplot

Construct the boxplot for student exam scores

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94



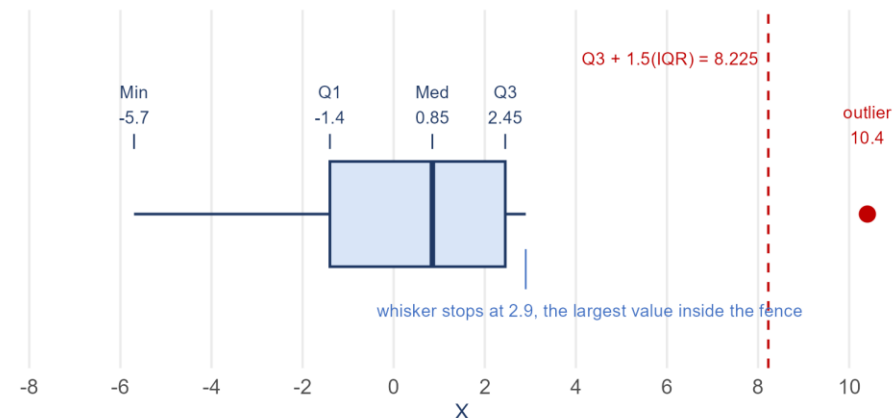
Ex2. Construct a Boxplot

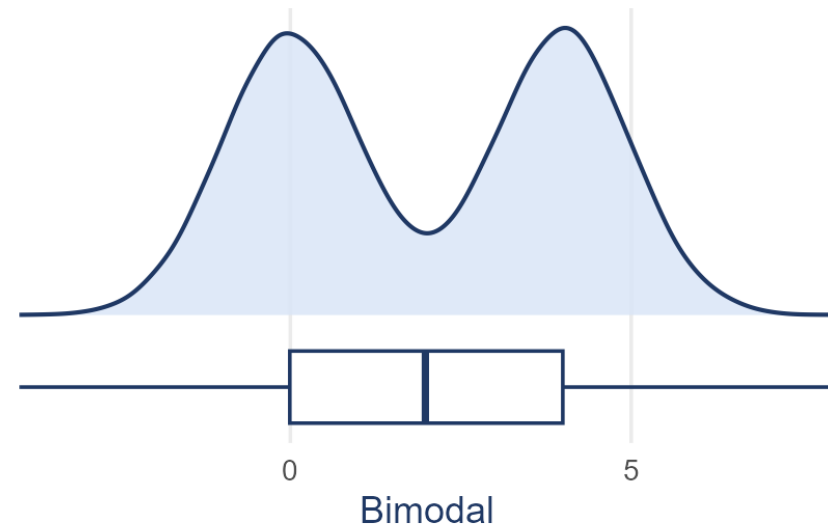
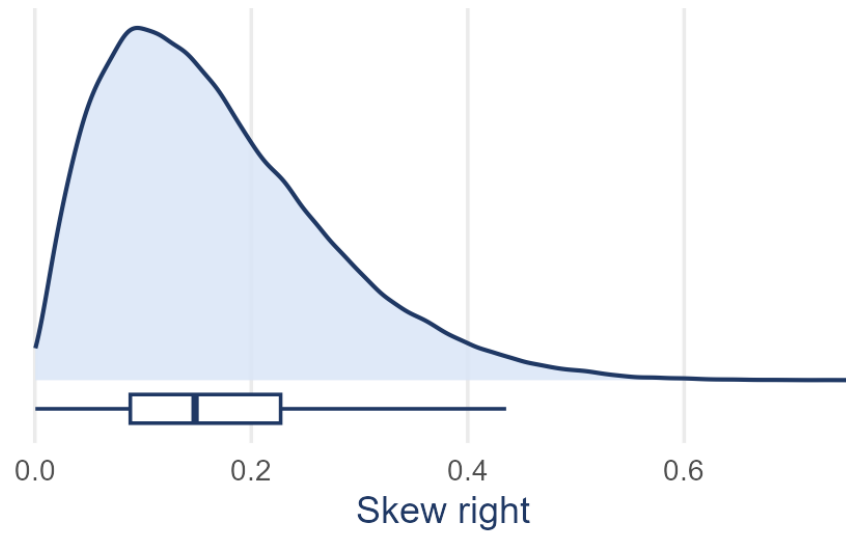
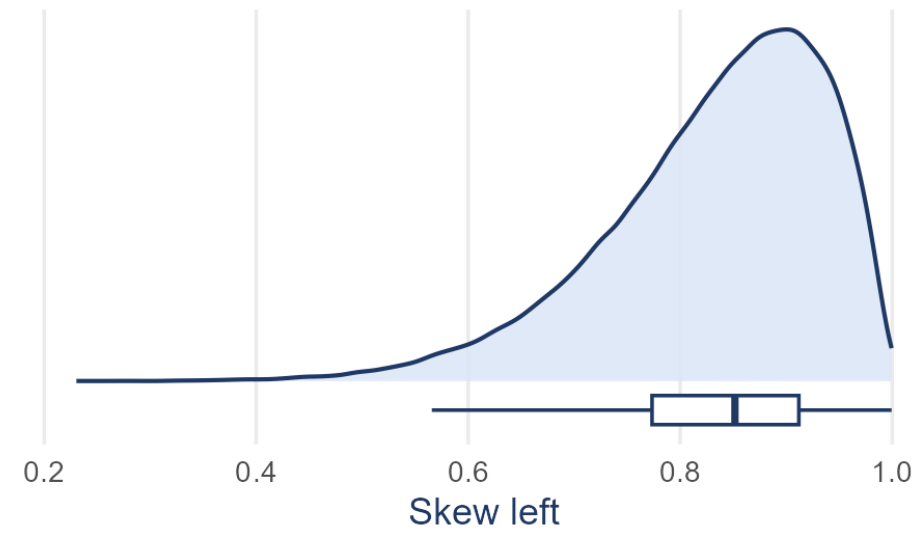
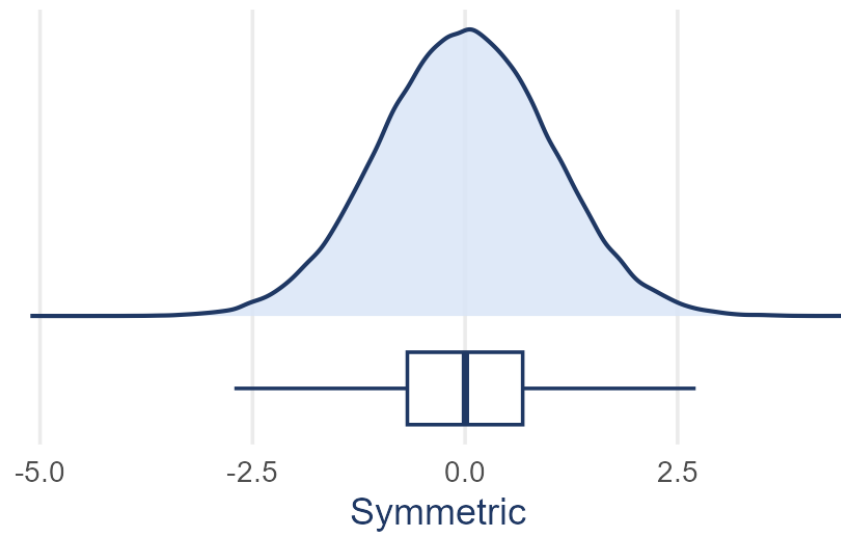
Consider the following 12 observations of a quantitative variable X

$X = \{-5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4\}$

Compute the 5 number summary and draw a boxplot

Compare the IQR with the range





Whisker lengths show skew. But the bimodal panel has the same boxplot as a symmetric one - a boxplot cannot show peaks or gaps, so read it next to a histogram.